

Clinical tooling for vascularization reconstruction on Contrast-Enhanced Micro CT 3D images.

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Preface

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Abstract

The accurate reconstruction of tissue vascularization in ex-vivo biopsies is essential to quantitative analysis of samples, to enable their comparison across various treatment profiles, and achieve the level of accuracy required for drug research. We aim to enable the analysis of antiangiogenic drugs for cancer treatment, specifically focused on pazopanib, by improving quantification of vascularization. Classical histology methods such as 2D histology suffer from a lack of accuracy due to the slicing process, and when used for vascularization reconstruction, lose the ability to reconstruct its structural nature. 3D techniques such as contrast enhanced Micro CT enables the acquisition of high resolution 3D images of ex-vivo samples aided by a contrast agent, but does not enable direct quantitative information extraction due to discontinuities, noise, low contrast and small features. In this document, a method for the structural reconstruction of vascularization from microCT images is presented, enabling analysis of the impact of antiangiogenic drugs on the vascularization of tumors in mice.

-> this can be improved and extended later

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Chapter 1

Introduction

When carrying out research, the process of data acquisition, processing and analysis requires expertise across a vast amount of diverse domains. It is natural for researchers to not have extensive knowledge across all fields required for a particular experiment, which is compensated for by relying on the existing corpus of prior research work. However replicability is a major challenge across all disciplines: prior research work often fails to be reproducible or extensible due to a lack of accessibility to the data acquisition hardware and methods, data processing tools, insufficiently documented procedures, or subjective biases resulting from prior experience of the person executing the process [1]. Free and open source tools palliate the data processing replicability issue, publication of the data as well as data versionning enable an understanding of the steps carried out modifying the data, and automation in processes such as segmentation remove the tacit data filtering done by a human in the annotation process.

When carrying out research in biology, an understanding of a target tissue is often required. Tissue microstructure is closely linked to its function: soft tissue analysis is foundational to modern medicine, being used everywhere from clinical environments to laboratories. Tissue samples are collected and imaged in order to gain an understanding of their structure and composition, called histology or histopathology in the case of diseased tissue. We use the term histology to refer to either interchangeably. Classically, this sampling and subsequent analysis has been done *ex-vivo* in 2D using a microscope, setting the gold standard for histology [2].

Histology is relevant in both clinical and laboratory environments. Clinically, histopathological analysis of tissue samples is extensively used for use cases such as cancer detection and grading [3] where a sample is analysed manually or with computer assistance. In laboratories for tasks such as clinical trials, histopathologists carry out manual microscopic evaluation [4]. Classical 2D histology presents the distinct disadvantage of being unable to obtain information at the same granularity in all three axis: slices may be stacked to obtain a 3D result, but resolution in the third axis is limited to section thickness, with the stacked planes separated by a dead space where no information is acquired [5], (adding a figure here for the loss in resolution is difficult to find?) and the cutting process induces changes in tissue structure [6].

All types of tissues can be analysed histologically. For the purpose of evaluating the use of the antiangiogenic drug Pazopanib, a tyrosine kinase inhibitor which blocks the creation of new blood vessels, as a means for cancer treatment, accurate 3d reconstruction of blood vessel vascularization is required due to simple volume estimates not capturing important parameters such as tortuosity, connectivity/branching and cross section profile. The shortcomings of classical 2D histology make the assessment and quantification of hollow and highly heterogeneous tissues like vasculature, for which spatial relationships are particularly relevant, challenging [5]. Due to these limitations alternatives such as 3D imaging methods that do not require slicing and have high spatial resolution in all axis are particularly relevant for blood vessel analysis.

To collect 3D data, the simplest is to extend 2D by stacking slices [7]. This method limits the resolution across the slicing axis, and does not alleviate the tissue deformation challenges. Non invasive 3D imaging methods rely on wavelengths able to further penetrate the tissue to be imaged: visible light can be used for shallow 3D imaging of soft tissues, with longer wavelengths such as x-rays able to penetrate completely through soft tissue as well as cartilage and bone. Although the machines and methods exist to collect high resolution 3D data using various wavelengths and associated techniques, such as μ MRI and μ CT, the analysis of high resolution 3D data is not as straightforward as with 2D histological slices: there are many more slices, often thousands, requiring software to assemble these into a 3D volume, and exponentially more time for manual annotation. Methods for segmenting in 3D lack standardization: there is high scan to scan variance between different tissue types, wide array of machines and machine settings, different contrast agents used at different concentrations and with different methodologies.

Our method of choice for analyzing blood vessels, μ CT, presents low contrast between soft tissue: techniques exist to increase contrast and improve visibility such as cryo-CT [8] or the utilization of contrast agents in contrast enhanced CT, called CECT. Although essential to increase the contrast between tissue types, these techniques still do not enable sufficient differentiation for small vascular networks such as those found in the murine tumors collected for the evaluation of Pazopanib to allow threshold simple based segmentation, as is commonly used for bone or samples with high contrast agent presence [5]. (should I link here to Lisa's thesis?)

Alternatives to threshold based segmentation exist, however due to subjectivity and annotation techniques, these suffer from strong variance between annotators [9]. This variance comes from multiple sources: expertise difference, tool type and experience, available time, subjectivity and more. This variance, combined with the time required to manually segment, means that small vascular networks remain difficult to obtain complete segmentations for, reducing the replicability of research and introducing noise into the data, increasing uncertainty margins.

Finally, when a segmentation can be obtained, it is often plagued by issues stemming from the nature of prior free segmentation techniques like threshold or non context aware methods: discontinuities in blood vessels, stemming from the lack of explicit vesselness constraints such as spatial continuity, mean that the final segmentation lacks the ability

to be evaluated based on higher level parameters like tortuosity, and instead are purely evaluated on volume (here again lisa? or her reference?).

In this thesis work will be carried out to create an open source, user friendly and robust small blood vessel extraction method for obtaining clinically relevant vasculature characteristics from μ CT data acquired in the context of the testing of Pazopanib, with a focus on ensuring re-usability and adoption by researchers. The pipeline will use 3DSlicer as a basis and take in and output data in portable data formats, to ensure re-usability of the work. Proven algorithms with user adjustable hyperparameters to enable easy re-use without requiring expertise in computer science are to be used. Implemented methods will be compared with existing pipelines for extracting fine blood vessels, and tested for robustness by examining performance on data previously considered non segmentable due to non uniform contrast agent staining.

Chapter 2

State of the art

2.1 Software in Research

The kinds of software used in a research setting vary enormously by location, as well as in scope, purpose, availability, expense and licensing. At the UCLouvain faculty IMMC (Institute of Mechanics, Materials, and Civil Engineering), research is carried out on biological tissue samples using computed tomography techniques. During this research, a variety of software is used: both as explicit steps in the pipeline for tasks such as segmenting samples in 3D, using Avizo or Dragonfly3D, or implicitly as elements of the pipeline that lay before the steps done by researchers (software running on CT machines) or in the infrastructure, such as the OS used on computing devices.

The guiding principles of research dictate that Open, reproducible, and replicable practices are a fundamental part of science, it is common however that scientific research be carried out in part or fully using closed source or proprietary software, with licenses for certain pieces of software reaching into the multiple thousands, such as is the case for Avizo by Thermo Fisher Scientific. Other pieces of paid software are part of the infrastructure such as the operating system Windows used throughout the lab, acting as an implicit cost to replication. These software are occasionally available under a “free for research” liencese, such as FreeD offered by Dragonfly3D [10], another piece of software used in the lab, discussed later. For most published research in biology, the relevant software to the research runs on top of a abstracted computational stack, and as a result we will focus on this “high level” software.

2.1.1 Software Licenses

Software licenses govern the terms under which a piece of software may be used, modified, and redistributed. They broadly fall into two categories: proprietary (closed source) and open source. Proprietary software restricts access to its source code and is generally distributed under a paid license, although exceptions exist in the form of freeware: proprietary software made available at no cost, such as Dragonfly3D’s FreED license. Open source software makes its source code publicly available and is in the vast majority of cases also free of charge, though commercial open source models exist such as Mozilla Firefox. Within open source licenses, meaningful distinctions exist between permissive licenses (such as MIT or BSD),

which place few restrictions on reuse, and copyleft licenses (such as the GPL family), which require that derivative works remain open source.

Open source software plays a particularly important role in scientific research for two reasons: First, being free removes a significant barrier to entry: any researcher, regardless of institutional resources, can access, run, and reproduce a pipeline built on open source tools. This is a prerequisite for replicability, one of the foundational principles of science. Secondly open source software is extensible: prior work can be built upon by accessing, studying and modifying its source code. This capacity for incremental improvement mirrors the broader scientific process itself, where each work builds upon prior results. An example of this is Orthanc [11], a medical image viewer whose architecture deliberately relies on pre-existing open components such as the Lua scripting language and standard networking protocols, enabling the community to extend its functionality without duplicating prior effort.

The standard for open and free software at the OS level is Linux, a kernel, on top of which various distributions are built such as Ubuntu as used in this thesis, or one of a variety of over one thousand others [12]. On the OS is run the software that is used during the various stages of scientific work. Of these stages, relevant to this thesis are the software required to process the outputs of a MicroCT machine: software able to reconstruct the 3D representation of the imaged target from the collection of 2D slices provided by the CT machine.

2.1.2 Software for 3D analysis

The outputs, after reconstruction, of a CT scan are in the form of slice-by-slice files. These 2D slice outputs are then combined into a 3D representation by dedicated software, of which many exist. Drawing from Dragonfly3D’s own comparative software list and a search for available tools, those relevant to this thesis can be divided along the same axis introduced in the previous section: commercial and open source or freeware. On the commercial side, the main options include (Bold indicating those actively used in the UCLouvain IMMC) **Dragonfly3D**, **Avizo**, **CTAn**, Amira, Analyze and CTVol (by Bruker), Image Pro, Imaris, MeVisLab, Mimics, ScanIP, Octopus, VG Studio, Zeiss Inspect. On the open source and freeware side, the principal options are 3D Slicer, Chimera, Blender, Dream3D, Drishti, ImageJ, FIJI, IMOD, MeshLab, OsiriX, ParaView, SimVascular, VesselKnife and VisIt.

Additionally, more specialised and focused tools, made in a research context exist such as SPROUT and nnU-Net, the latter being a deep learning framework for medical image segmentation. For the purposes of this thesis, candidate software was evaluated against a set of technical criteria, ordered by importance:

1. Import a 3D scan from individual 2D slices in standard formats such as TIFF, PNG, JPEG, BMP, or DICOM.
2. Export data to non-proprietary formats to ensure data portability
3. Allow the development and implementation of plugins
4. Ability to (manually or automatically) segment parts of a scan, especially of blood vessels and availability of external tools specifically designed for vascular segmentation.

Due to the goal of developing software for practical real world use, the following aspects were also weighed:

1. Availability of dedicated support forums & tutorial videos
2. Active development
3. Prior use by the lab researchers, and by researchers of the field more broadly.
4. Ease of installation and barrier to entry

Due to these criteria, especially the practical aspects, standalone software such as DeepVesselNet [13] and SPROUT [14] were discarded as candidates and focus was set to comparing the commercial tools actively in use in the laboratory for blood vessel segmentation (Avizo, Dragonfly3D) to open source alternatives 3D Slicer and ImageJ/FIJI. ImageJ and its distribution FIJI have a rich history in biological image analysis with an extensive plugin ecosystem relevant for this work, such as the Frangi vesselness [15] algorithm, but their handling of large 3D volumes and native support for vascular network extraction is limited. 3D Slicer [16] emerged as the most suitable tool for this thesis, due to its pre-existing plugin intended specifically for advanced vessel segmentation: The Vascular Modeling Toolkit [17] Beyond satisfying the core technical criteria of importing 3D scans as stacks of 2D slices, as well as import and export of open formats such as DICOM, 3D slicer also has extensive instructions and prior work in manual and semi-automatic segmentation. The Extension Manager is particularly relevant to this work: the proprietary pieces of software studied also had available plugins/extensions, however these are usually expensive. For 3D Slicer, the VMTK (the Vascular Modelling Toolkit) mentioned above, with its open code enabling modification, enables working on an existing proven basis. The recent project R-Vessel-X [18]: especially developed for the segmentation of blood vessel trees from medical images, was also retained as relevant.

The combination of a stable, widely adopted platform with active community support and specialised vascular tooling makes 3D Slicer the most appropriate choice for extracting and analysing the vasculature from CECT μ CT data in a reproducible and extensible manner.

Problem 1. *In an extensive and diverse software landscape, users stick to what is familiar and easy to use, regardless of the impact on scientific rigor. Our software must thus be based on an open platform: 3DSlicer, be easy to use: click only installation and use, and have ample, clear documentation.*

2.2 Tissue imaging

Tissue imaging is a critical step in both clinical and research contexts, where it is used to acquire a better understanding of the organism being examined than is possible with indirect measurements. Tissue imaging methods can be characterized by multiple factors, of relevance in this document: dimensionality (1D, 2D, 3D, 4D) and invasiveness (ex-vivo / in-vivo). Ex-vivo imaging requires the removal (and generally destruction or alteration) of the tissue to be imaged, whereas in-vivo can be done in a live organism, offering the distinct advantage of being able to measure the same tissue at multiple points in time.

Tissue imaging enables the extraction of parameters that are qualitative and/or quantitative: in clinical and research settings, it is common to extract simple qualitative measurements, such as a difference in size, due to it both being easier and quantification not being required. However, for tasks such as comparing the impact of a drug on a structure, quantification becomes relevant: Understanding the three-dimensional structure of biological tissue is a prerequisite for the analysis of structure-altering drugs such as Pazopanib, where changes in tumor vascularization are inherently spatial and quantifiable across multiple parameters. Classical 2D histology remains the gold standard for analysis of biological tissues, valued for its high discriminative power, the wide range of available stains, and its compatibility with immunohistochemistry [5]. Despite this, it suffers from fundamental limitations when the target of analysis is a spatially complex 3D structure. Physical sectioning of the sample is destructive, is not orientation-agnostic, and introduces deformation artifacts that are difficult to compensate even with embedding techniques [19] (Here could use a greet source?). Samples undergo structural changes over time during preparation: they dry out, and certain elements oxydize, although techniques exist to mitigate this [5].

To achieve 3D imaging from 2D histology, 2D slices are stacked across a virtual axis [6]. The resolution achievable when stacking 2D slices to reconstruct a virtual third dimension is limited by slice thickness and inter-slice registration errors. Certain optical techniques making use of slices allow partial recovery of 3D information from 2D acquisitions, such as confocal microscopy, light sheet microscopy, and optical coherence tomography [8] (is it good to reference a secondary source like this?), they are limited in sample penetration depth and volume. When the target structure to be imaged and understood does not follow a single preferred axis, as is fundamentally the case for vascular networks and especially those of tumors, the limitations of slice-based histology make proper reliable quantification of blood vessel parameters impossible.

2.2.1 3D imaging techniques for histology

Non-destructive 3D volumetric imaging methods overcome many of the limitations of 2D slice based histology by collecting data uniformly across dimensions, enabling virtual slicing across any plane without requiring physically sectioning the sample. Techniques for 3D imaging range from the aforementioned microscopy techniques using visible or near infrared light, to techniques utilizing magnetic fields such as magnetic resonance imaging (MRI) or X-Ray imaging in the form of X-ray microfocus computed tomography (CT). These digital techniques allows qualitative and quantitative 3D microstructural analysis of tissues and of

their constituents: analysis is not restricted to a single orientation and does not require sample destruction. They also have associated high resolution variants more commonly used for research, allowing micrometer level imaging: μ MRI and μ CT, with μ CT reaching higher resolutions (smaller voxel sizes). These techniques are the most relevant for ex-vivo tissue histology at the scale of small animal models. μ MRI provides fully resolved 3D images non-invasively and non-destructively, and has an inherently high-contrast for soft tissues, though resolution is limited to the range of tens of micrometers. MicroCT reaches higher resolutions, but suffers from low contrast when imaging soft tissue, when extra techniques are not used [2].

2.2.2 Contrast-enhanced μ CT

As mentioned, standard X-ray microfocus computed tomography (microCT or μ CT) suffers from low contrast between soft tissues making the visualization of blood vessels difficult. To improve contrast two common approaches exist: imaging methodology changes, and sample modification. Imaging techniques such as phase-contrast microCT (PC-CT) utilize the refractive properties of the X-rays rather than its absorption alone, enhance soft tissue edge detection at the cost of higher complexity and long acquisition times (**Do I need to source this if it's already discussed in the cryo source that follows?**). Tissue modifying techniques that aim to increase contrast include the use of various Contrast Enhancing agents, termed CECT. For the tumor vascularization studied in this thesis, ex-vivo tissue binding CESAs are used, namely Hf-WD POM due to its nondestructive nature and low tissue shrinkage during incubation after excision. Techniques exist that can be used in combination with CESAs: contrast can be increased using freezing to increase contrast Cryogenic contrast-enhanced microCT (cryo-CECT) which preserves tissue microstructure with reduced deformation [8].

CECT is of particular relevance for imaging large vascular networks because of the ability to inject the CE agent into the vasculature [20], in tumors however the microvasculature present a challenge: small blood vessels do not lend themselves to perfusion, so alternatives are used namely diffusion. Tumors present also a particularity in that they are frequently partially or entirely devoid of residual hemoglobin, preventing or reducing the action of CESAs and thereby reducing contrast. Diffusion based contrast-enhancing agents present disadvantages in terms of complexity when used for imaging: the diffusion of CE agents throughout the tissue happens from the outside in, resulting in a gradient of the amount of CESA and as a result a gradient in contrast, as opposed to perfusion CESAs that follow the blood vessels. CECT was utilized for the acquisition of the tumor images in this thesis due to its ability to enable differentiation of the blood vessel boundaries and reasonable image acquisition times.

Problem 2. *The wide diversity of available modalities for 3D imaging: μ MRI and μ CT, with their subtypes: cryo-CECT, phase-contrast CT, Contrast-Enhanced CT, the different machines and their acquisition parameters, variability in samples and their preparation: fixation methods, staining agents and methods, staining duration, tissue types, presents the challenge of creating a method that is re-usable, even across different tissues within the same*

lab. Any method used for the segmentation of small blood vessels should be robust to the gradients caused by diffusion CECT, and able to handle variable contrast levels.

2.3 Information extraction from images

The purpose of imaging a tissue is to generate a fixed, deterministic representation at a given point in time. After imaging, information is obtained from the data using one of a variety of processing methods. Classically, this information extraction was carried out by trained professionals: for medical image analysis, an expert in cancerous tissues is capable of identifying and evaluating qualitative and quantitative metrics for a given sample. This process of information extraction can be seen as a form of data compression: only the relevant conclusions from the image are kept and represented in multiple fashions: a simple aggregated conclusion of the form “cancer” or “no cancer”, a classification of areas in the image into a certain class or type.

With the introduction of computing to the world of bio and medical imaging, higher levels of representation were also enabled, where the output of a pipeline may be the reconstruction of a structured output such as a network or skeleton, as is seen with software skeletonization methods [21]. Computers lend themselves well to information extraction and processing, especially compression. Different techniques or algorithms exist to extract information of interest from an image, ranging from simple classical methods such as the OTSU method for edge detection [22] to more computationally expensive, complex and inscrutable deep learning. Deep learning algorithms are known to act as a form of information compression algorithm [23], starting from a data source with many individual data points, such as an image, and ending with as little as a single class output as illustrated in [24].

2.3.1 Methods of Segmentation

Image segmentation methods span a wide spectrum of complexity, from classical signal processing approaches to modern deep learning architectures, and vary in output format, from pixel wise annotation and single class prediction to higher-order structural extraction methods that reason about the topology of extracted objects as in skeletonization.

At the simplest end, intensity-based thresholding methods such as Otsu’s method [22] operate by iteratively finding an optimal intensity threshold that separates foreground from background by minimizing intra-class variance. Such methods are computationally inexpensive and interpretable, but are sensitive to noise, imaging artifacts, and intensity inhomogeneities, as they optimize for a numerical goal. This is particularly problematic in the context of μ CT imaging, where algorithms that reason globally over an entire image can fail due to gradients across the image from phenomena such as beam hardening, or as mentioned previously when contrast enhancing agents are used, the diffusion gradient. Region-growing and watershed algorithms extend this idea by incorporating spatial connectivity, iteratively expanding labeled regions from seed points according to local intensity gradients, making them more robust to global intensity variation, however remaining susceptible to over- or under-segmentation in complex structures, and creating discontinuities due to not integrating priors of the structure they are segmenting, such as the connectivity of blood vessels.

Classical machine learning approaches introduced feature engineering as an intermediate

step: handcrafted descriptors such as Gabor filters, Hessian-based vesselness filters or particularly relevant: Frangi’s multiscale vessel enhancement filter [25]. The Frangi vesselness algorithm has been particularly influential in vessel segmentation tasks, as it explicitly models the prior of tubular geometry inherent in blood vessels. These methods offer greater robustness than simple thresholding but depend heavily on the quality of the engineered features, requiring user fine tuning of hyper parameters to optimize performance for a given domain, and as a result of this feature engineering, fail to generalize across imaging conditions.

2.3.2 Machine learning in biology

Forms of Machine Learning

As mentioned, machine learning relies on the process of iterative optimization, and can be as simple as optimizing for an intensity threshold. Deep learning methods, and in particular the paradigm of fully convolutional encoder-decoder architectures such as U-Net [26], represent a major revolution in the way image segmentation was done, and offered a leap in performance. U-Net presented breakthrough performance in bio and medical imaging due to its ability to segment large images with high compute and data efficiency, and across multiple scales by introducing skip connections between encoder and decoder pathways. These allow the network to combine low-level spatial detail with high-level semantic context, enabling the segmentation of thin structures that require context such as cells and, as relevant here, capillaries. However as mentioned previously, 2D imaging and by extension 2D segmentation present limitations for blood vessels; variants of U-Net extending into the third dimension exist to palliate this, such as 3D U-Net [27].

Transformer-based architectures have come to the forefront of predictive performance by their more general computation model: they remove the locality prior of convolutions, and therefore have the ability to model long-range spatial dependencies. Transformers function as learned compression functions where the encoder progressively distills the input volume into a compact latent representation encoding the most task-relevant features, which the decoder then maps back into a dense prediction [23]. Due to their more general structure lacking the prior of locality, basic transformers require more data to train than basic convolutional networks, complicating their use in medical imaging due to the associated costs and barriers to data collection, although techniques such as pre-training can alleviate this need [28].

Classification techniques

In order of granularity, the most basic form of classification consists in classifying an image as a whole. This form of machine learning applied to medical images consists in outputting a single unified class given an input piece of data, such as classifying an x-ray of a bone as “broken” or “not broken”. Its simplicity and ease of collecting training data means classification models formed the first widely used machine learning models [29]. Segmentation offers more granularity by classifying regions (most commonly pixels) in an image into classes. Segmentation plays a central role in accelerating and standardizing medical image analysis by enabling quantitative analysis of different kinds: by classifying pixels with semantic labels (semantic segmentation), partitioning of objects (instance segmentation) or a combination of both across the entire image space (panoptic segmentation) [30]. In the context of volumetric

biological imaging, segmentation extends naturally into three dimensions, with the paradigm of convolutional neural networks also being extended into the third dimension by making use of 3D convolutions in the place of 2D.

Beyond classification and pixel or voxel-wise labeling, methods exist that aim to extract higher-order structural constructs. For tree and network-like structures such as vascular networks, skeletonization algorithms abstract a binary segmentation mask to a centerline representation that preserves the branching topology of the original structure. In order to have centerline extraction be an output of a deep learning model, it can be formulated directly as an optimization problem alongside pixel or voxel wise segmentation, enabling it to be utilized in the training loop as will be explored later. At the highest level of abstraction, graph-based representations encode the vascular network as a set of nodes (bifurcation points and endpoints) connected by edges (vessel segments), each annotated with attributes such as radius, length, and tortuosity. Abstract representations enable the highest quality downstream quantitative analysis, including branching order, fractal dimension, or inter-vessel spacing, impossible to derive from a raw segmentation mask alone [13], as well as enable simulations with tools such as [31]

Structural extraction

Moving between levels of representation modifies the balance between information fidelity, robustness to segmentation errors, and the specificity of the downstream analysis task. Segmentation masks carry spatial information in the image frame while being difficult to analyze quantitatively, limiting to the use of lower order metrics such as volume. Graphs on the other hand are the richest representations quantitatively, enabling the extraction of volume as well as self reflexive parameters such as tortuosity. Generally pipelines chain these representations sequentially due to simplicity and scrutability: it is easier to collect and annotate data in image space than in graph space, and graph space data does not necessarily translate perfectly back to image space.

Sequential pipelines have the weakness that errors propagate and compound across each stage, and can result in more fragile pipelines, sensitive to noise or changes in domain. Disconnections in the segmentation mask can produce disconnected graphs if the algorithm does not account for this possibility, and if proper care has not been given to provide hyperparameters that enable tuning, or the ability to modify the code used at the different steps, the pipeline becomes unusable. This motivates approaches that integrate structural priors directly into the segmentation model, or that formulate centerline extraction as a joint optimization rather than a post-processing step (**Problem 3.1**).

Loss functions

Machine learning models are optimized by minimizing a loss function that quantifies the discrepancy between model predictions and ground truth annotations. The choice of loss function central to the performance of a model as it implicitly encodes priors about the structure of the problem. Types of prediction errors are weighted differently and the geometric or topological properties that predictions should satisfy change. For usecases such as vascular segmentation it is particularly consequential due to the severe class imbalance,

or difference in amount of the “target class” vessel pixels when compared to the “background class”, as well as for connectedness.

Loss functions such as cross-entropy used for binary classification calculate the loss on a pixel by pixel basis, based on the predicted distribution. It can be used for segmentation [26], however it is important to recall that our prediction task carries a heavy imbalance due to many more pixels being the background class. With cross-entropy loss, each prediction is independent, meaning for our biased distribution there is a prior of predicting background and producing fragmented or incomplete vessel predictions. There exist loss functions created that integrate the class imbalance explicitly, such as [32] however these do not integrate the prior of connectivity or any spatial relatedness: each prediction is treated as independent.

For blood vessels specifically, breaks in connectedness can be difficult to reconnect downstream, and purpose made loss functions such as center-lineDice (clDice) [33] exist. clDice makes use of the aforementioned skeletonization techniques, that integrate the prior of connectedness, and build it into the loss function for predictions. Using clDice loss for a model thus encodes the prior that blood vessels are connected tubular structures into the model structure itself, avoiding the need for separate skeletonization and repair steps.

Optimizing for even higher order parameters is also possible, such as clinically relevant parameters. CF-Loss [34] optimizes the loss function for retinal multi-class vessel segmentation and vascular feature measurement, focusing on improving the performance on clinically relevant metrics. clDice and CF-Loss can be used in conjunction with other loss functions such as the lower order DICS, and act as a complimentary extra constraint for network training.

The selection of a loss function or combination of functions through weighing amounts to choosing which structural priors to encode in the optimization objective. For our case of vascular segmentation in small tumor μ CT volumes with heavy class imbalance and the goal of extracting clinically relevant metrics that are topological correct, a compound loss will be used that combines the imbalance prior, as seen in DICE, and a structurally oriented loss function (**Problem 3.2.**).

Problem 3.1. *Methods for extracting blood vessels can do so at different levels of abstraction. It is important to be able to select the level of abstraction desired in the software, and for any method used to export to have adjustable parameters for this purpose.*

Problem 3.2. *The selected evaluation method should encode the structure of blood vessels as a prior in some form, in order to ensure that quantified prediction performance matches with qualified performance as experienced by the user.*

2.4 Imaging and Segmentation of vasculature

The segmentation and reconstruction of vascular networks from 3D data is a long-standing and active problem in medical image analysis across multiple tissue types, organisms and even taxa (**Here trying to say it's animals, plants etc**), with applications ranging from brain imaging [35] to oncology, as discussed here. Blood vessels present a distinctive and consistent set of geometric priors despite their diversity: they are tubular, connected, branching structures and contain blood which can provide a different imaging profile to surrounding tissue. The challenge of their size, spatial spread and contrast is compounded in the context of tumor microvasculature due to the small size of the murine tumors collected, and blood vessel sizes reaching the lower limits of μ CT resolution as explored in [36]. Vessels are also poorly contrasted as a result of this sizing.

The first line of approaches to separating vessels from surrounding tissue in the context of CECT imaging is that of threshold segmentation [36]: due to the presence of a contrast agent, the vasculature network of interest is able to be mostly segmented using only a process involving defining all voxels within a certain intensity range as vessels. This is a form of extraction in the imaging space, providing no structural information, although in situations with high contrast and well connected networks, can approximate a higher order extraction.

The dominant classical algorithms developed for vessel extraction are based around the prior of tubeness, and expanded upon by the Frangi vesselness filter [25], and more recently improved in [37]. By analyzing the eigenvalues of the Hessian matrix of local image intensities at multiple Gaussian scales, a vesselness score is produced for each voxel that responds to tubular structures while suppressing circular and planar structures. The Frangi filter is popular enough to have earned an implementation in ImageJ [15]. However, the filter's response degrades at vessel bifurcations, at the endpoints of vessels, and in regions of low contrast, and it has 4 hyper parameters requiring manual tuning. [37] proposed extensions to the vesselness formulation that improve responses at junctions and at low contrast boundaries to increase robustness of the filter.

Beyond filter-based mathematically driven approaches that work on small image localities lay methods relying on optimization: minimal path methods formulate vessel centerline extraction as an optimization problem. The path of least cost is taken between user-defined endpoints, with cost derived from vesselness or image intensity. [38] incorporate the geometry of vessels adding radius as an additional dimension of the path space. This class of methods is particularly well-suited to expert-in-the-loop approaches where small amounts of data need to be annotated: a user can place seed points that guide the extraction of a complete vascular tree, requiring no training data, and enabling iterative improvement. This strength is also a weakness: the low scalability means that for a densely vascularized tumor, or a tumor with non uniform characteristics, many seed points can be required which is impractical. Automatic seed point placement remains an open problem.

2.4.1 Deep learning for vessel extraction

Deep learning has been applied to vascular segmentation in for blood vessel extraction with notable success: DeepVesselNet [13] introduced a family of architectures designed for vessel segmentation, centerline prediction, and bifurcation detection in 3D angiographic data by making explicit use of the structural priors inherent to blood vessels as secondary learning targets. Training on centerline and bifurcation labels jointly with segmentation masks improves topological results when compared to simple segmentation training, by providing extra prior generating signals to the model, highlighting the benefits of explicitly utilizing structural priors. V-Net [39] extended U-Net [26] into 3D and improved performance by modifying the loss function; Dice loss is used for the overlap between predicted and ground truth segmentation masks as opposed to cross-entropy, as it is better suited for the class-imbalanced setting characteristic of vessel segmentation where vessel voxels represent a small minority of the image volume.

A major and difficult to circumvent limitation of (supervised) deep learning approaches for vascular extraction is their dependence on relatively large volumes of annotated training data. Manual annotation of vascular networks in μ CT images is time intensive and requires expert knowledge, and the resulting annotations are known to exhibit significant inter-annotator variability [9]. This variability represents an irreducible uncertainty in the ground truth that places an upper bound on the performance achievable by any supervised method, especially those training on voxel level data. Work-arounds exist, such as removing the inter-annotator noise by first performing structural extraction on the data, and then using these structures to segment the base image, however this presents the problem of introducing the biases of the skeletonization step backwards in the pipeline, into the deep learning model weights. Additionally, models trained on one imaging protocol, contrast agent, sample batch, or tissue type exhibit degraded performance when applied to data from a different distribution: this is termed domain shift, and is particularly acute in methods with imaging methods containing many adjustable parameters and machine specificities like μ CT imaging where the aforementioned parameters vary widely between laboratories (**Problem 4.2**).

2.4.2 Existing pipelines

Several software tools and pipelines have been developed to support vascular segmentation and analysis workflows. TKTubeTK [40] offers a library of algorithms for tubular structure segmentation and graph extraction built on the ITK framework. The Vascular Modeling Toolkit (VMTK) [17] integrates in 3DSlicer and provides a user friendly comprehensive suite of tools for vascular segmentation and vascular extraction: centerline extraction, surface reconstruction, and hemodynamic simulation. The recent VesselKnife [41] provides an integrated pipeline targeting vessel segmentation, skeletonization, and graph extraction from μ CT data. For analysis, SKAN [21] provides a well documented Python library for the quantitative analysis of skeleton graphs extracted from binary segmentation masks, enabling computation of branch length distributions, tortuosity, and network connectivity. The SimVascular [31] software extends vascular extraction into simulation, enabling downstream modeling of blood flow within reconstructed vascular geometries.

The landscape of vascular segmentation tools reflects a broader conflict throughout bio-informatics, medical informatics and bioimagery: general-purpose deep learning segmentation frameworks offer good performance on the datasets they are trained on, but require large annotated datasets and offer limited interpretability or controllability as well as generally basic existing implementations into existing tools; classical model-based methods such as Frangi filtering are more transparent and adjustable, built into existing tools and easy to use, but require manual parameter tuning and struggle with complex vascular geometries. Methods that explicitly encode the structural priors of vascular networks: connectivity, vessel geometry and branching topology represent a middle ground that remains underutilized in laboratory settings, particularly for the specific setting of small tumor μ CT imaging with small amounts of data and variable imaging conditions (**Problem 4.1**, **Problem 4.2**).

Problem 4.1. *Current methods for segmentation often ignore the structural priors that underly the data generation process. An effective, robust and transferrable method for blood vessel segmentation must thus encode the relevant structural priors, namely connectedness, shape, and branching structure*

Problem 4.2. *Deep learning based segmentation methods that rely on a large corpus of images fit to the distribution on which they are trained, that suffer when the domain shifts due to changes in imaging methodology or sample variance, are not robust enough, nor transferrable enough to enable replicable work across multiple datasets. Any algorithm to extract blood vessels from tumors must contain user adjustable hyper parameters to enable fitting to the parameters of their data, as well as avoid the need for training data beyond a single test example.*

Problem 4.3. *Existing vascular segmentation pipelines are typically designed and validated for a single imaging modality, contrast strategy, or tissue type. The fragmentative nature of research means there exist few user friendly modular pipelines with adjustable parameters that can be adapted to the specific imaging characteristics of small tumor μ CT (non uniform contrast agent diffusion, limited sample size, low contrast and tumor vessel specificities). Robust, reproducible vascular reconstruction usable for quantifying the vasculature network with an accuracy sufficient for clinical and research applications is thus out of reach for most lab work . A robust pipeline must expose interpretable parameters that allow adaption to the users specific imaging context, without requiring retraining as with common deep learning methods, or re-annotation.*

Chapter 3

Problem statement

As discussed in the 7 problem points laid out previously, our principal challenge lays in delivering a work that satisfies three distinct actors:

1. Any users of the program, for whom computing is a tool that existing amongst others used during their work, and where tool ease of use and polyvalence is essential;
2. Scientists the program for research, to whom performance, speed of attaining results, repeatability and adjustability to their exact modalities and needs are essential
3. The scientific community, for whom replicability, re-usability and ease of use are essential in being able to extend any paper they may want to build upon;

Requirements amongst these groups overlap, and from these user profiles, as well as the 7 problem points, we obtain the following problem statement:

How may a software pipeline for the segmentation and extraction of blood vessel networks from CECT images be designed, such that it is accessible and usable by domain scientists such as biologists, for whom software is not a primary tool (**Problem 1**), where the software is robust to intensity gradients and variable contrast levels of CECT images, without requiring dataset-specific retraining or re-annotation (**Problem 2, 4.2, 4.3**), capable of producing outputs in easily portable binary masks with interpretable, adjustable parameters (**Problem 3.1**), that extracts and is evaluated using metrics that encode the priors of vascular networks as opposed to only voxel level metrics (**Problem 3.2, Problem 4.1**)?

3.1 Problem resolution approach

The proposed approach to solving the problem laid out above is as follows:

1. A user centric approach, where the end user is placed in the segmentation loop and offers feedback to the algorithm, through placement of anchors [38]
2. A method that makes use of existing, extendable software 3DSlicer [16]
3. A method that integrates voxel level metrics such as DICE [32] and vasculature relevant metrics such as [33] and [34]
4. A method that is well documented, and enables reproducibility
5. A method that outputs a portable format of segmentation, namely voxel level segmentation, as individual slices or a DICOM imaging format

3. PROBLEM STATEMENT

The approach will make use of the ability of 3DSlicer to visualize the outputs of a plugin immediately, and include the user by means of allowing adjustment of the segmentation through anchor points and hyperparameters for the segmentation reconnection process. The segmentation will begin with a simple window thresholding segmentation done by the user or automatically based on the anchor points selected. The algorithm will then attempt to reconnect and grow the segmentation based on its loss functions, region growing and other lower order techniques, with the integration of the priors of vascularization by means of the loss functions used. The user will then be able to export the segmentations into a portable format.

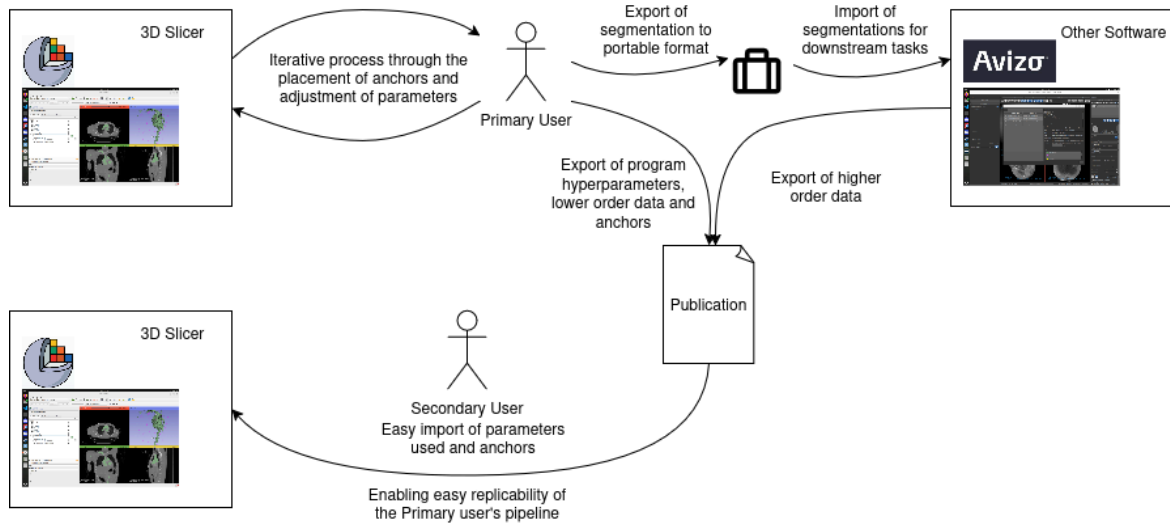


Figure 3.1: Preliminary synthesized diagram for the annotation of vascularization and sharing of results

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